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10 / 10 submitted

Q1. Find smallest and largest number

CODE

```
arr = list(map(int, input().split()))
smallest = min(arr)
largest = max(arr)
print("Smallest number:", smallest)
print("Largest number:", largest)
```

INPUT

12 5 18 3 25 9

OUTPUT

(no output)

Q2. Missing Number

CODE

```
def missing_number(arr):
    n = len(arr) + 1
    expected_sum = n * (n + 1) // 2
    actual_sum = sum(arr)
    return expected_sum - actual_sum

arr = list(map(int, input().split()))
print(missing_number(arr))
```

INPUT

4 2 1 6 5

OUTPUT

(no output)

Q3. Second largest number

CODE

```
nums = list(map(int, input().split()))

largest = float('-inf')
second_largest = float('-inf')

for x in nums:
    if x > largest:
        second_largest = largest
        largest = x
    elif x != largest and x > second_largest:
        second_largest = x

print(int(second_largest))
```

INPUT

12 5 18 3 25 9

OUTPUT

(no output)

Q4. Most repeated number

CODE

```
arr = list(map(int, input().split()))

freq = {}
for x in arr:
    freq[x] = freq.get(x, 0) + 1

max_count = max(freq.values())
most_repeated = min([k for k, v in freq.items() if v == max_count]) # if tie, smallest

print(most_repeated)
```

INPUT

2 4 6 2 5 3 1 2

OUTPUT

(no output)

Q5. Unique elements

CODE

```
def unique_elements(arr):
    result = []
    seen = set()
    for x in arr:
        if x not in seen:
            seen.add(x)
            result.append(x)
    return result

# INPUT: 2 4 6 2 5 6 7
arr = list(map(int, input().split()))
print(*unique_elements(arr))
```

INPUT

2 4 6 2 5 6 7

OUTPUT

(no output)

Q6. Rotate an array by 'N' positions

CODE

```
def rotate_array(arr, n):
    n = n % len(arr)          # handle n ≥ length
    return arr[n:] + arr[:n]

# Example
arr = [2, 5, 6, 1, 3, 4]
n = 2
print(*rotate_array(arr, n))
```

INPUT

2 5 6 1 3 4

OUTPUT

(no output)

Q7. Remove all duplicates in an array

CODE

```
arr = list(map(int, input().split()))

seen = set()
result = []

for x in arr:
    if x not in seen:
        seen.add(x)
        result.append(x)

print(*result)
```

INPUT

2 4 6 2 7 8 3 4

OUTPUT

(no output)

Q8. Print all 0's followed by non-zero without changing the order

CODE

```
arr = list(map(int, input().split()))
zeros = [x for x in arr if x == 0]
nonzeros = [x for x in arr if x != 0]
result = zeros + nonzeros
print(*result)
```

INPUT

1 0 4 0 2 5 0 7 0 6 0

OUTPUT

(no output)

Q9. count vowels, consonants, digits, and spaces

CODE

```
sentence = input()

vowels = 0
consonants = 0
digits = 0
spaces = 0

for ch in sentence:
    if ch == ' ':
        spaces += 1
    elif ch.isdigit():
        digits += 1
    elif ch.isalpha():
        if ch.lower() in "aeiou":
            vowels += 1
        else:
            consonants += 1

print("Vowels:" + str(vowels))
print("Consonants:" + str(consonants))
print("Digits:" + str(digits))
print("Spaces:" + str(spaces))
```

INPUT

Hello World 123

OUTPUT

(no output)

Q10. Palindrome

CODE

```
s = input().strip()
if s == s[::-1]:
    print(f"{s} is a palindrome")
else:
    print(f"{s} is not a palindrome")
```

INPUT

saveetha

OUTPUT

(no output)